

Kubernetes Cluster Certificate Expiry Runbook

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Category	kubernetes	Severity	P1 - Critical
Domain	Container Orchestration	Est. Duration	45-90 minutes
Tags	kubernetes, certificates, tls, expiry, kubeadm, apiserver, etcd		

Overview

Expired Kubernetes certificates cause the API server, etcd, and kubelets to reject connections, making the cluster completely unmanageable. kubeadm-provisioned clusters have 1-year certificate validity. This runbook covers detection, renewal via kubeadm, and verification. Act before expiry — renewal requires API access.

Prerequisites

- SSH access to all control plane nodes
- kubeadm installed on control plane nodes
- Root/sudo access on control plane nodes
- Backup of /etc/kubernetes/pki directory

Response Steps

Step 1: Check Certificate Expiry Dates

Check all cluster certificates and identify which ones have expired or will expire soon.

```
kubeadm certs check-expiration  
  
# Manual check for specific cert:  
openssl x509 -in /etc/kubernetes/pki/apiserver.crt -noout -enddate
```

NOTE: Certificates expiring within 30 days should be renewed proactively during a maintenance window.

Step 2: Backup Existing Certificates

Before renewing, backup all existing certificates and kubeconfig files.

```
cp -r /etc/kubernetes/pki /etc/kubernetes/pki.bak.$(date +%Y%m%d)  
cp /etc/kubernetes/admin.conf /etc/kubernetes/admin.conf.bak  
ls -la /etc/kubernetes/pki.bak.*
```

Step 3: Renew All Certificates

Use kubeadm to renew all certificates at once. This updates all certs in /etc/kubernetes/pki.

```
kubeadm certs renew all  
kubeadm certs check-expiration
```

NOTE: Renewed certs will not take effect until the API server and controller-manager are restarted.

Step 4: Restart Control Plane Components

The API server and controller-manager read certs at startup. Restart them to load the new certs.

```
# For kubeadm static pods, move and restore manifests to force restart:  
mv /etc/kubernetes/manifests/kube-apiserver.yaml /tmp/
```

```
sleep 20
mv /tmp/kube-apiserver.yaml /etc/kubernetes/manifests/
sleep 30 && kubectl get nodes
```

Step 5: Update kubeconfig Files

The admin.conf kubeconfig contains the embedded client certificate. Regenerate it.

```
kubeadm init phase kubeconfig admin
cp /etc/kubernetes/admin.conf ~/.kube/config
kubectl cluster-info
```

Step 6: Restart Kubelet on All Nodes

Each node's kubelet uses its own certificate to communicate with the API server. Restart kubelet on all nodes.

```
# On each node (control plane + workers):
systemctl restart kubelet
systemctl status kubelet
kubectl get nodes
```

NOTE: SUCCESS: All nodes return to Ready state and kubectl commands work normally.

Rollback Steps

Rollback Step 1: Restore Backup Certificates

If renewal caused issues, restore the backed-up certificates.

```
cp -r /etc/kubernetes/pki.bak. /etc/kubernetes/pki
systemctl restart kubelet
```

Rollback Step 2: Emergency API Server Access

If the API server is unreachable, access etcd directly to diagnose.

```
ETCDCTL_API=3 etcdctl --endpoints=https://127.0.0.1:2379
--cacert=/etc/kubernetes/pki/etcd/ca.crt --cert=/etc/kubernetes/pki/etcd/server.crt
--key=/etc/kubernetes/pki/etcd/server.key member list
```

Step Dependencies

Step	Depends On	Can Parallelize With
Step 1: Check Expiry	—	—
Step 2: Backup Certs	Step 1	—
Step 3: Renew All Certs	Step 2	—
Step 4: Restart Control Plane	Step 3	—
Step 5: Update kubeconfig	Step 3	Step 4
Step 6: Restart Kubelet	Step 4, Step 5	—